

1.6 Bipartite graphs

Let $r \geq 2$ be an integer. A graph $G = (V, E)$ is called *r-partite* if V admits a partition into r classes such that every edge has its ends in different classes: vertices in the same partition class must not be adjacent. Instead of ‘2-partite’ one usually says *bipartite*.

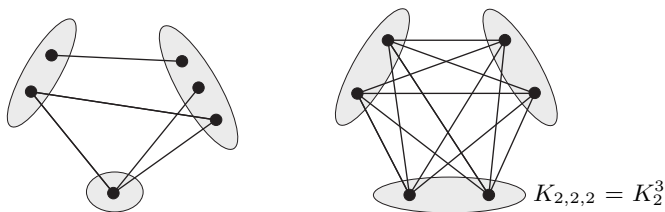
*r-partite**bipartite*

Fig. 1.6.1. Two 3-partite graphs

An r -partite graph in which every two vertices from different partition classes are adjacent is called *complete*; the complete r -partite graphs for all r together are the *complete multipartite* graphs. The complete r -partite graph $\overline{K^{n_1}} * \dots * \overline{K^{n_r}}$ is denoted by $K_{n_1, \dots, n_r}$; if $n_1 = \dots = n_r =: s$, we abbreviate this to K_s^r . Thus, K_s^r is the complete r -partite graph in which every partition class contains exactly s vertices.⁶ (Figure 1.6.1 shows the example of the octahedron K_2^3 ; compare its drawing with that in Figure 1.4.3.) Graphs of the form $K_{1,n}$ are called *stars*; the vertex in the singleton partition class of this $K_{1,n}$ is the star’s *centre*.

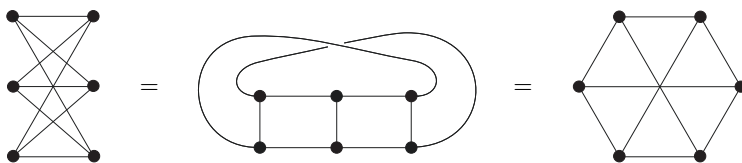
*complete
r-partite* $K_{n_1, \dots, n_r}$
 K_s^r *star
centre*

Fig. 1.6.2. Three drawings of the bipartite graph $K_{3,3} = K_3^2$

Clearly, a bipartite graph cannot contain an *odd cycle*, a cycle of odd length. In fact, the bipartite graphs are characterized by this property:

odd cycle

Proposition 1.6.1. *A graph is bipartite if and only if it contains no odd cycle.*

[1.9.4]
[5.3.1]
[6.4.2]

⁶ Note that we obtain a K_s^r if we replace each vertex of a K^r by an independent s -set; our notation of K_s^r is intended to hint at this connection.

(1.5.1) *Proof.* Let $G = (V, E)$ be a graph without odd cycles; we show that G is bipartite. Clearly a graph is bipartite if all its components are bipartite or trivial, so we may assume that G is connected. Let T be a spanning tree in G , pick a root $r \in T$, and denote the associated tree-order on V by $\leq_T$. For each $v \in V$, the unique path rTv has odd or even length. This defines a bipartition of V ; we show that G is bipartite with this partition.

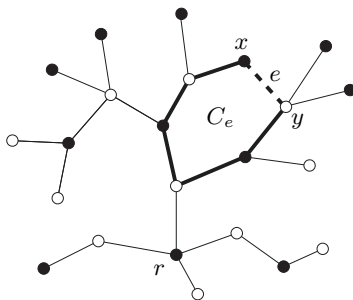


Fig. 1.6.3. The cycle C_e in $T + e$

Let $e = xy$ be an edge of G . If $e \in T$, with $x <_T y$ say, then $rTy = rTxy$ and so x and y lie in different partition classes. If $e \notin T$ then $C_e := xTy + e$ is a cycle (Fig. 1.6.3), and by the case treated already the vertices along xTy alternate between the two classes. Since C_e is even by assumption, x and y again lie in different classes. $\square$

1.7 Contraction and minors

In Section 1.1 we saw two fundamental containment relations between graphs: the ‘subgraph’ relation, and the ‘induced subgraph’ relation. In this section we meet two more: the ‘minor’ relation, and the ‘topological minor’ relation. Let X be a fixed graph.

A *subdivision* of X is, informally, any graph obtained from X by ‘subdividing’ some or all of its edges by drawing new vertices on those edges. In other words, we replace some edges of X with new paths between their ends, so that none of these paths has an inner vertex in $V(X)$ or on another new path. When G is a subdivision of X , we also say that G is a *TX*.⁷ The original vertices of X are the *branch vertices* of the *TX*; its new vertices are called *subdividing vertices*. Note that

subdivision
TX of X

branch
vertices

⁷ The ‘ T ’ stands for ‘topological’. Although, formally, *TX* denotes a whole class of graphs, the class of all subdivisions of X , it is customary to use the expression as indicated to refer to an arbitrary member of that class.